

Nxcleus.



Describe an internal process. Get it running *inside your walls.*

A plain-language description becomes a **planned, certified, fleet-built, adversarially-QA'd** process — running forever on AMD hardware you control.

built on amd mi300x · rocm + vllm

one external call per build · zero at runtime

live at nxcleus.tech

the problem

The most regulated companies are the *least able* to use AI.

pain 01

Data cannot leave

Banks, insurers, clinics and law firms are forbidden — by regulators and their own policies — from sending customer data to external AI clouds. Frontier assistants are simply banned inside the walls.

pain 02

Processes run by hand

KYC checks, claims triage, regulatory reporting: high-volume work still runs on consultants, spreadsheets and months-long software projects that are obsolete on arrival.

pain 03

The best tools can't come inside

A frontier harness like Claude Code could do the work — policy keeps it outside. Local models alone lack the harness to carry a real process end-to-end. The gap is architecture, not model weights.

\$206^B

spent yearly on financial-crime compliance operations alone, most of it labor

lexisnexis · true cost of compliance

6–18^{mo}

typical delivery time for one internally-built process automation

enterprise it benchmarks

0

frontier AI tools approved for raw customer data in a typical regulated enterprise

the wedge we build on

the architecture

One designed-in crossing. *Everything else stays.*



external · 1 call

The planner your pick

A frontier model — e.g. GPT-5.6 or Claude Fable 5 — sees **only** a sanitized brief, governed by your own confidentiality policy. It designs the work topology and the model bill-of-materials. It never touches raw data.

the boundary

local · everything else

Your AMD hardware

Intake, certification against the raw context the frontier never saw, the parallel build fleet, consolidation, adversarial QA — and **every operate-phase run**. All on customer-controlled AMD compute.

Sovereign Mode rebinds even planning on-fleet — **zero external calls**. Any egress during a sovereign job raises a violation and paints the run red.

egress ledger · per-job receipt

planner · gpt-5.6 / sanitized

certifier · glm-4.6 · sealed

coder pool · qwen · sealed

oracle · gemma · sealed

1 crossing · rest on mi300x

live & proven

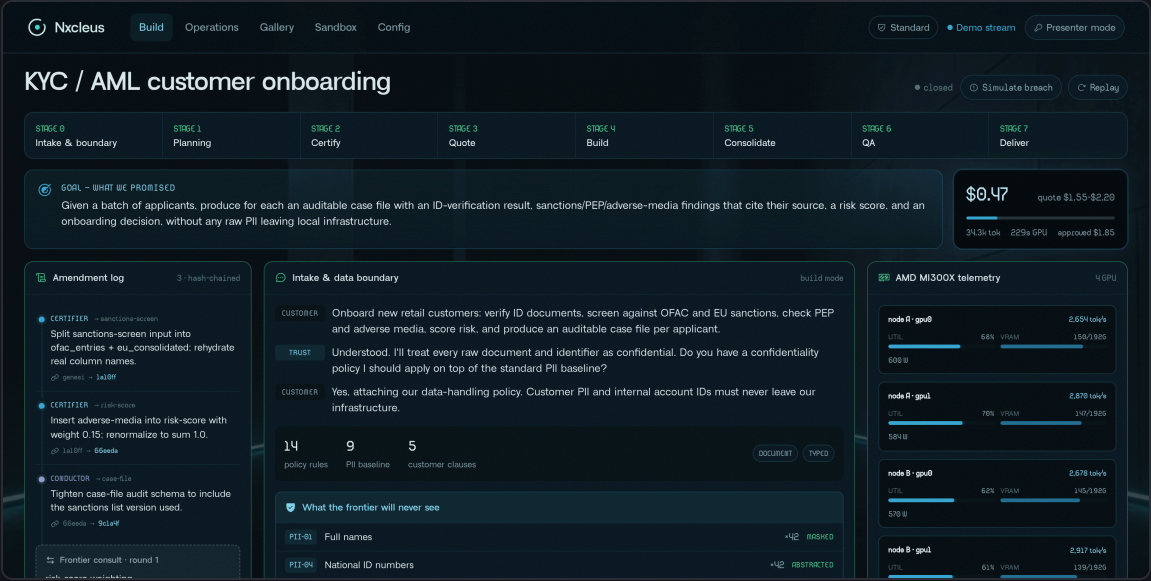
Working today. *Not a mock-up.*

Full live build, stages 0 → 7. A KYC / AML onboarding process was planned, certified, fleet-built, QA'd and registered in a single session on the MI300X fleet.

136 of 138 planted patterns caught. The insurer sandbox seeds deliberately hard anomalies — identities split across records, invisible to regex or rule engines. Catching them takes reasoning, not string-matching.

The proof streams live. Per-node rocm-smi utilization, VRAM, tokens/sec and the egress ledger render in the cockpit as the build runs. The AMD fleet is visible, not asserted.

Try it yourself. Build cockpit, operations registry, judge sandbox with three synthetic companies, and the Sovereign Mode toggle — live at nxcleus.tech.



the build cockpit: stages 0→7, goal anchor, amd mi300x telemetry & live egress ledger

kyc 0→7 live on mi300x 136/138 caught event-sourced · replayable

under the hood

The build pipeline — *stages 0 → 7.*

0

local

Intake & boundary

Prompt, files, code, DBs + your policy. A trust model strips what's forbidden, composes the sanitized brief.

1

external

Planning

The frontier planner sees **only the brief**; designs the work topology + model bill-of-materials. The one crossing.

2

local

Certify & goal

A strong local model completes the plan against the real raw context, logs a signed amendment trail, pins the goal.

3

quote

Quote

The BoM becomes a deterministic, itemized price — before a single build token is spent. Approve to proceed.

4

local fleet

Parallel build

Capability-routed coder agents build in waves with declared touch points; a conductor reviews between waves.

5

local

Consolidation

A consolidator merges wave outputs, gated by objective tests — nothing lands unless it passes.

6

local

Adversarial QA

Inspectors attack the build; a lineage-independent Numeric Oracle re-derives the math; the goal is re-checked.

7

local

Delivery

The verified process lands in the registry with docs, tests & a metered invoice. Then it runs forever.

the model

A *lifecycle*, not a factory.



A factory ships a deliverable and walks away. **Nxcleus builds a process and then keeps it alive** — owned, running, and refinable inside your walls.

phase 01 · once

Build

Describe → sanitize → plan → certify against the raw context → build in parallel waves → adversarial QA → the process lands in the operations registry.

phase 02 · forever

Operate

The certified process runs on new data, fully local — **zero frontier calls** — metered per run, with oracle spot-checks as a live warranty.

phase 03 · on demand

Refine

Change requests re-open planning under the same triage model; every refinement ships a versioned, diffed, re-certified **v(n+1)**. Old versions stay runnable.

the loop build → operate → refine → operate...

every process becomes an owned, versioned asset

built on amd

Inside the boundary, every token runs on *AMD silicon*.

running today

The live demo serves on a **dedicated MI300X node** — 192 GB HBM3, vLLM on ROCm — with **AMD-hosted serverless MI300X** capacity for burst. Every local token runs on AMD silicon; the map below is the single-tenant scale target.

node a · 1 gpu

**Gemma-4-31B · Gemma-4-26B ·
Qwen3.6-35B**

Numeric oracle, trust layer / OCR, adversarial inspector. Gemma is merit-picked: deliberately lineage-independent from the coder fleet.

node b · 5 gpus · bf16

GLM-4.6 — the local brain

Certifier, wave conductor, consolidator, Sovereign-Mode planner. Allocation justified line-by-line by the plan's model bill-of-materials.

nodes c-d · 2 gpus

**Qwen3-Coder-Next · Qwen3.6-27B ·
Devstral**

Capability-routed coder pool — each model to what it is measurably best at.

external · 1 call

The stage-1 frontier planner — **your pick, e.g. GPT-5.6 (via OpenRouter) or Claude Fable 5**, or bring your own keys. Sees the sanitized brief, nothing else. Rebindable per seat; removed entirely in Sovereign Mode.

1x

MI300X node live today · 192 GB HBM3 · vLLM on ROCm · live rocm-smi telemetry

8x

MI300X single-tenant scale profile · 1.5 TB pooled HBM3

1

external call per job. All other inference is AMD-hosted

the economics

Frontier intelligence is **capex**, not marginal cost.

**\$0.54**

to build — one frontier call, ever

\$0.009

per applicant to run — zero frontier calls

\$100k+

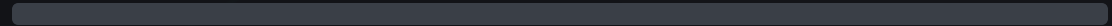
and months — a consultancy building the same process once, decaying from day one, data still leaving

The live KYC / AML process, measured at nxcleus.tech/operations. **Runs meter at 3–5× over cost** and still land under a nickel per applicant; pilot builds price at \$2–10k against ~\$1.26 of measured compute. Software margins, priced from live meters.

the market

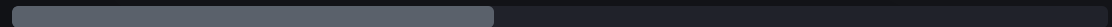
Where external AI is *banned*.

t a m \$46B by 2030



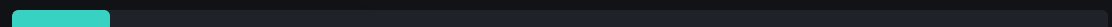
Intelligent process automation, global — growing ≈21% a year.

s a m \$20B by 2030



Automation spend inside regulated industries: banking, insurance, healthcare, legal.

s o m \$120M year 3



Sovereign AI automation for mid-size regulated firms. CIS, EU & MENA beachhead.

Banks

kyc · aml · surveillance

Insurers

claims · underwriting

Fintech

onboarding · risk

Healthcare

records · phi

Legal

discovery · review

"We operate *where external AI is banned*."



defensibility

What didn't exist before — and why it's *hard to copy*.

01 The boundary architecture

Customer-policy-governed sanitization, with a local certifier that checks the plan against data the frontier never saw. An architecture, not a feature toggle.

03 The goal anchor

Delivery is verified against the customer's original ask, not the plan's drift. Trust that copilots can't retrofit.

05 The toolsmith moat

Where a step is deterministic, models forge their own scripts, fitted to the business's exact schemas — certified runs get faster, cheaper, more repeatable.

02 Triage authority

The frontier authors; a local model finishes and certifies — with a visible, signed amendment log. The small model corrects the big one.

04 Capability-aware routing

Every model tagged by what it's measurably best at — the routing decision visible per module, GPU allocation justified by the plan's bill-of-materials.

06 Compounding switching costs

Every delivered process lands in the registry as a versioned, refinable asset. Leaving means abandoning a growing library.

what's next

now

A working MVP on a live URL: full 0→7 builds, egress ledger, judge sandbox, Sovereign Mode. Every local seat on AMD MI300X.

h2 2026

Two paid design-partner pilots in regulated firms; security posture & audit packaging; the sovereign appliance spec.

2027

A sovereign AMD appliance offering and a cross-customer library of certified process templates.

A hundred processes in every enterprise, waiting to run *inside the walls*.

nxcleus.tech

oxys studio

Yarasul Bychok · lead

Ansar Kazbek



scan to try it live